

Proof. **1. Reduction to the case $a, b > 0$.** If $a = 0$ or $b = 0$ then the left-hand side is 0 and the right-hand side is non-negative, so the inequality holds. Hence we may assume $a > 0$, $b > 0$.

2. Introduce the conjugate exponent. Let $p > 1$ and define q by

$$\frac{1}{p} + \frac{1}{q} = 1 \quad \implies \quad q = \frac{p}{p-1} > 1.$$

3. Convexity of the exponential function. The function $\varphi(t) = e^t$ is convex on $\mathbb{R}$. By Jensen's inequality, for any $\lambda \in [0, 1]$ and any real numbers x, y ,

$$e^{\lambda x + (1-\lambda)y} \leq \lambda e^x + (1-\lambda)e^y. \quad (1)$$

4. Choice of the convex combination. Set

$$\lambda = \frac{1}{p}, \quad 1 - \lambda = \frac{1}{q}, \quad x = p \ln a, \quad y = q \ln b.$$

Then

$$\lambda x + (1-\lambda)y = \frac{1}{p} p \ln a + \frac{1}{q} q \ln b \quad (1)$$

$$= \ln a + \ln b \quad (2)$$

$$= \ln(ab). \quad (2)$$

5. Apply Jensen's inequality. Substituting the choices of λ, x, y and using (2) in (1) gives

$$e^{\ln(ab)} \leq \frac{1}{p} e^{p \ln a} + \frac{1}{q} e^{q \ln b}. \quad (3)$$

Since $e^{\ln(ab)} = ab$ and $e^{p \ln a} = a^p$, $e^{q \ln b} = b^q$, we obtain

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}. \quad (3)$$

6. Extension to the non-negative case. Both sides of (3) are continuous in (a, b) on $[0, \infty)^2$; therefore the inequality, proved for all $a, b > 0$, remains valid when either $a = 0$ or $b = 0$.

7. Equality condition. Equality in Jensen's inequality (1) occurs exactly when the two arguments are equal, i.e. when $x = y$. With the chosen x, y this means

$$p \ln a = q \ln b \iff a^p = b^q.$$

Hence equality in Young's inequality holds iff $a^p = b^q$.

□